

TE Curve Verification Pipeline Familywise ONNX Report - residual_harmonic_lstm_sequence_dense360

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates exported ONNX models from the dataset input-mode retraining program. Each dataset/input-mode section uses dataset-matched held-out test curves and lists the exact model artifacts loaded from `models/`.

Rank-3 temporal ONNX exports are evaluated on the sequence-window test contract stored in each `training_config.snapshot.yaml`, including `sequence_length`, `sequence_stride`, `sequence_target_position`, and `maximum_sequences_per_curve`. The collage pages keep the measured TE trace at the original full-curve resolution; temporal ONNX predictions are overlaid at the evaluated sequence-target angles.

The report is diagnostic and family-specific. It does not replace an official multi-index model-promotion decision.

Output Artifacts

- output directory:
`output/validation_checks/track2_familywise_onnx_report/residual_harmonic_lstm_sequence_dense360/2026-07-10-12-52-29_track2_residual_harmonic_lstm_sequence_dense360_familywise_onnx_report` ;
- summary YAML:
`output/validation_checks/track2_familywise_onnx_report/residual_harmonic_lstm_sequence_dense360/2026-07-10-12-52-29_track2_residual_harmonic_lstm_sequence_dense360_familywise_onnx_report/track2_familywise_onnx_report_summary.yaml` ;
- model inventory CSV:
`output/validation_checks/track2_familywise_onnx_report/residual_harmonic_lstm_sequence_dense360/2026-07-10-12-52-29_track2_residual_harmonic_lstm_sequence_dense360_familywise_onnx_report/model_inventory.csv` ;
- per-curve metrics CSV:
`output/validation_checks/track2_familywise_onnx_report/residual_harmonic_lstm_sequence_dense360/2026-07-10-12-52-29_track2_residual_harmonic_lstm_sequence_dense360_familywise_onnx_report/per_curve_metrics.csv` .

Simplified Dataset + Setpoints

- dataset: `simplified_dataset` ;
- input mode: `setpoints` ;
- evaluated family: `residual_harmonic_lstm_sequence_dense360` ;
- dataset root: `data/simplified_dataset` .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	<code>te_residual_harmonic_lstm_sequence_dense360_fw_simplified_setpoints</code>	<code>2026-07-10-04-03-58_te_residual_harmonic_lstm_sequence_dense360_fw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
backward	<code>te_residual_harmonic_lstm_sequence_dense360_bw_simplified_setpoints</code>	<code>2026-07-10-04-23-03_te_residual_harmonic_lstm_sequence_dense360_bw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
global	<code>te_residual_harmonic_lstm_sequence_dense360_global_simplified_setpoints</code>	<code>2026-07-10-03-42-49_te_residual_harmonic_lstm_sequence_dense360_global_simplified_setpoints</code>	<code>simplified_curve_v1</code>

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	<code>models/simplified_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense360/forward/2026-07-10-04-03-58_te_residual_harmonic_lstm_sequence_dense360_fw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense360/forward/2026-07-10-04-03-58_te_residual_harmonic_lstm_sequence_dense360_fw_simplified_setpoints/python/residual_harmonic_lstm_sequence-epoch=047-val_mae=0.00358223.ckpt</code>
backward	<code>models/simplified_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense360/backward/2026-07-10-04-23-03_te_residual_harmonic_lstm_sequence_dense360_bw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense360/backward/2026-07-10-04-23-03_te_residual_harmonic_lstm_sequence_dense360_bw_simplified_setpoints/python/residual_harmonic_lstm_sequence-epoch=030-val_mae=0.00360440.ckpt</code>
global	<code>models/simplified_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense360/global/2026-07-10-03-42-49_te_residual_harmonic_lstm_sequence_dense360_global_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense360/global/2026-07-10-03-42-49_te_residual_harmonic_lstm_sequence_dense360_global_simplified_setpoints/python/residual_harmonic_lstm_sequence-epoch=055-val_mae=0.00360012.ckpt</code>

Aggregate Metrics

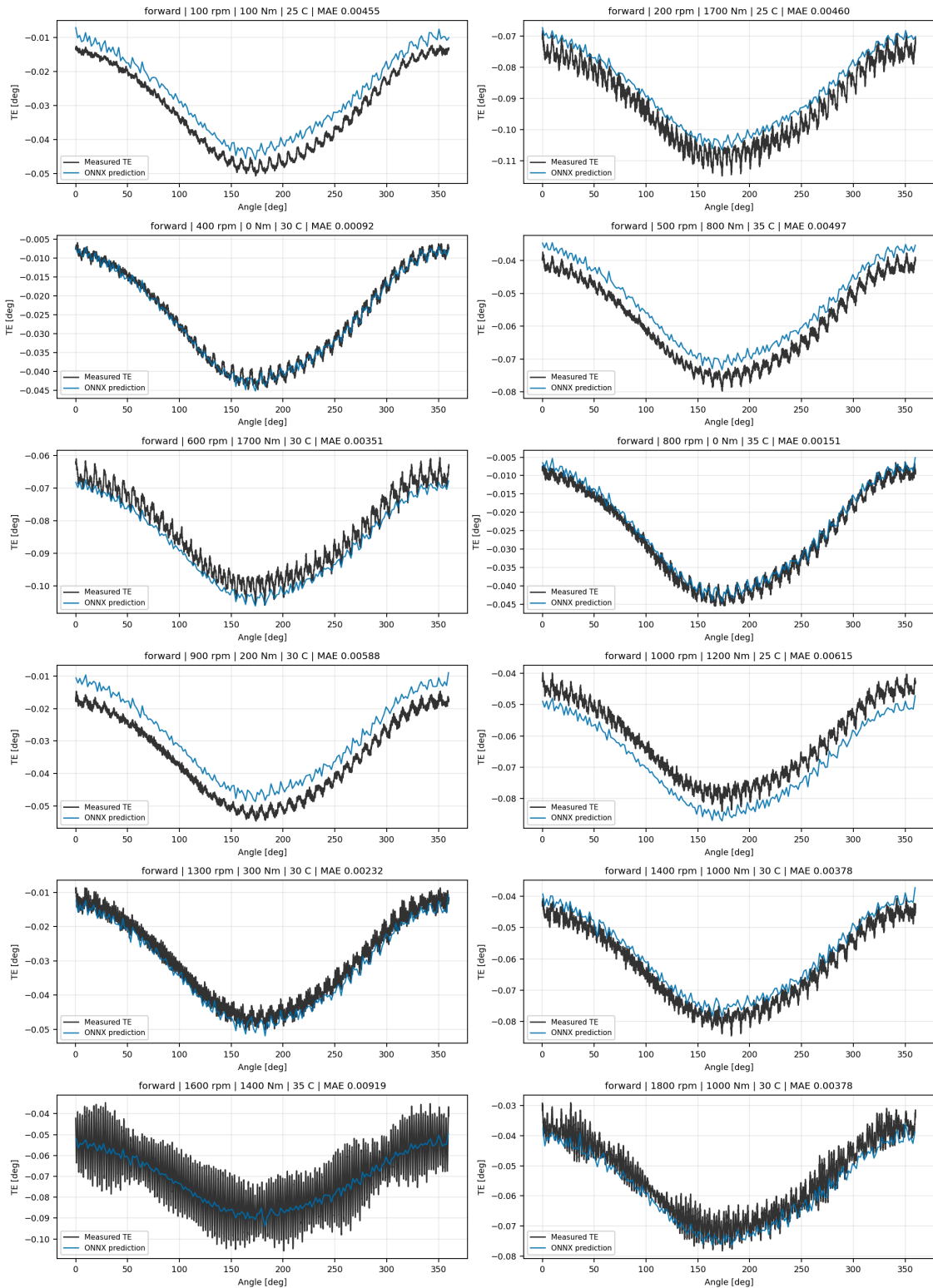
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	97	0.003324	0.003680	7.770	14.041
backward	97	0.003636	0.004035	8.436	13.508
global	194	0.003443	0.003807	8.022	14.410

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000286	0.002861	0.001494	0.003143
backward	-0.000432	0.002975	0.001785	0.004322
global	0.000167	0.002885	0.001594	0.003801

Forward 12-Curve Page

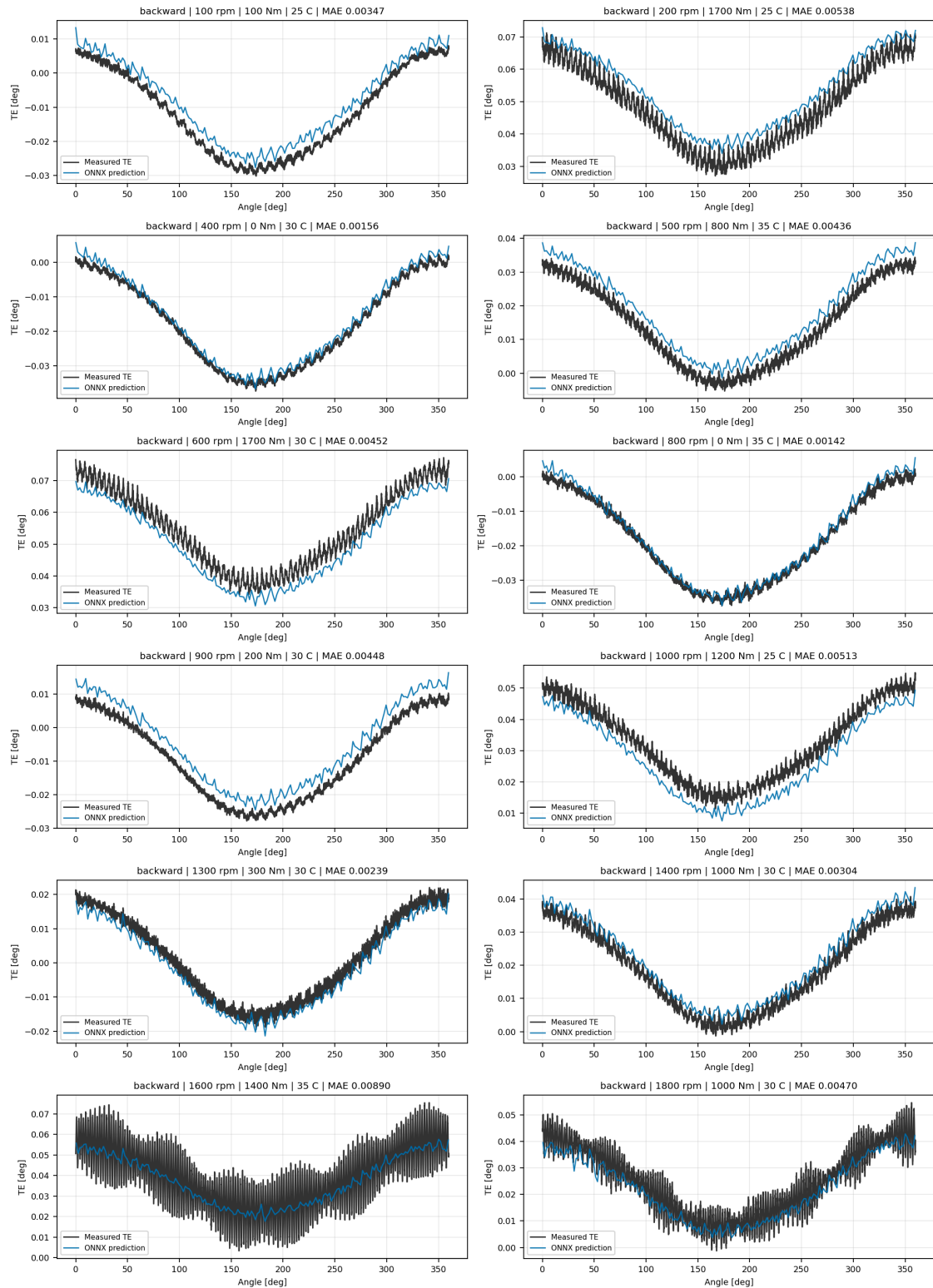
Simplified Dataset + Setpoints | residual_harmonic_lstm_sequence_dense360 | forward



simplified_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

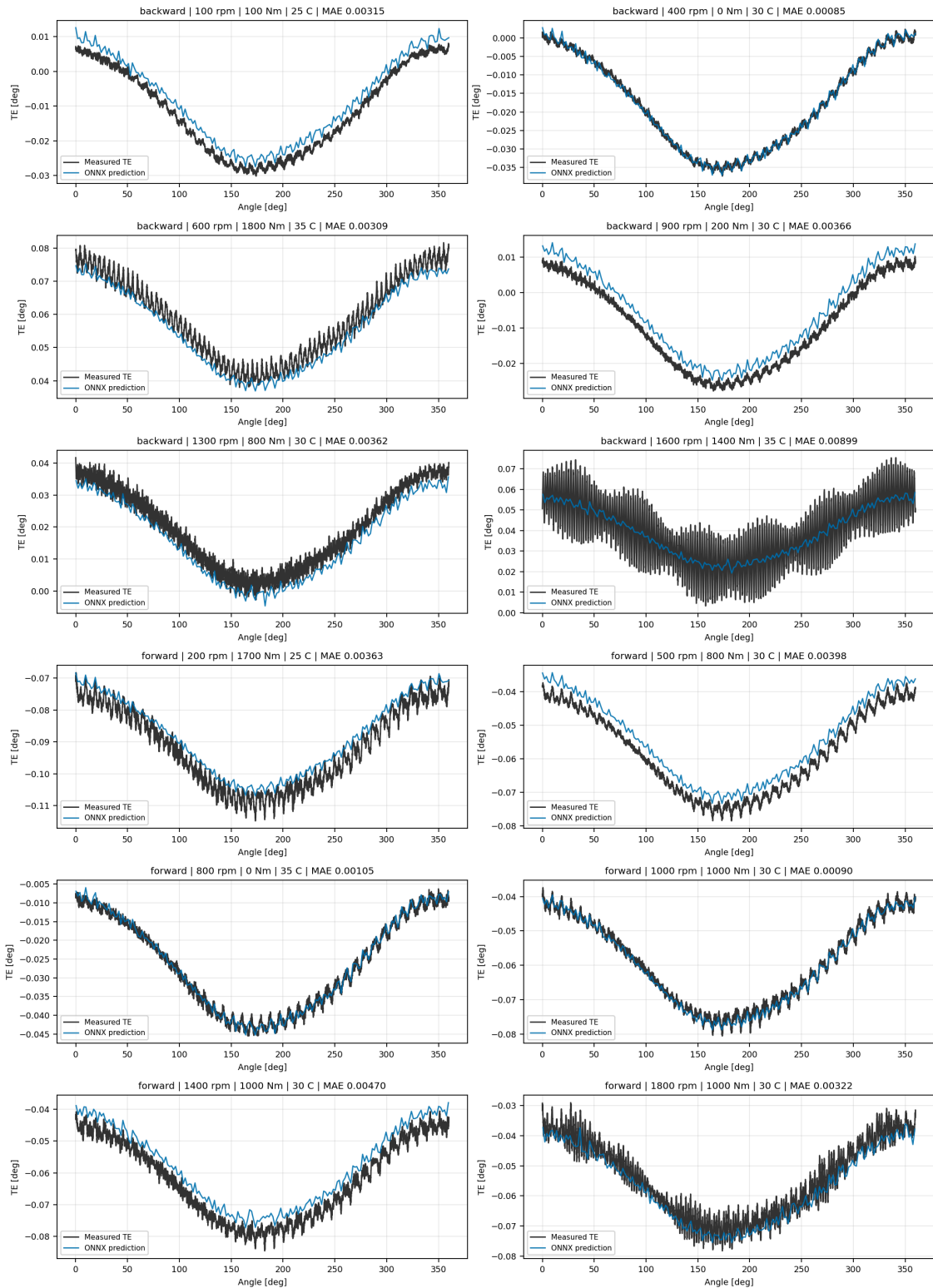
Simplified Dataset + Setpoints | residual_harmonic_lstm_sequence_dense360 | backward



simplified_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Simplified Dataset + Setpoints | residual_harmonic_lstm_sequence_dense360 | global



simplified_dataset_setpoints global 12-curve collage

Polished Dataset + Setpoints

- dataset: polished_dataset ;
- input mode: setpoints ;
- evaluated family: residual_harmonic_lstm_sequence_dense360 ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_residual_harmonic_lstm_sequence_dense360_fw_polished_setpoints	2026-07-10-05-27-52_te_residual_harmonic_lstm_sequence_dense360_fw_polished_setpoints	polished_setpoint_curve_v1
backward	te_residual_harmonic_lstm_sequence_dense360_bw_polished_setpoints	2026-07-10-05-51-16_te_residual_harmonic_lstm_sequence_dense360_bw_polished_setpoints	polished_setpoint_curve_v1
global	te_residual_harmonic_lstm_sequence_dense360_global_polished_setpoints	2026-07-10-04-53-57_te_residual_harmonic_lstm_sequence_dense360_global_polished_setpoints	polished_setpoint_curve_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense360/forward/2026-07-10-05-27-52_te_residual_harmonic_lstm_sequence_dense360_fw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense360/forward/2026-07-10-05-27-52_te_residual_harmonic_lstm_sequence_dense360_fw_polished_setpoints/python/residual_harmonic_lstm_sequence-epoch=089-val_mae=0.00201259.ckpt
backward	models/polished_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense360/backward/2026-07-10-05-51-16_te_residual_harmonic_lstm_sequence_dense360_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense360/backward/2026-07-10-05-51-16_te_residual_harmonic_lstm_sequence_dense360_bw_polished_setpoints/python/residual_harmonic_lstm_sequence-epoch=102-val_mae=0.00199077.ckpt
global	models/polished_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense360/global/2026-07-10-04-53-57_te_residual_harmonic_lstm_sequence_dense360_global_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense360/global/2026-07-10-04-53-57_te_residual_harmonic_lstm_sequence_dense360_global_polished_setpoints/python/residual_harmonic_lstm_sequence-epoch=101-val_mae=0.00197679.ckpt

Aggregate Metrics

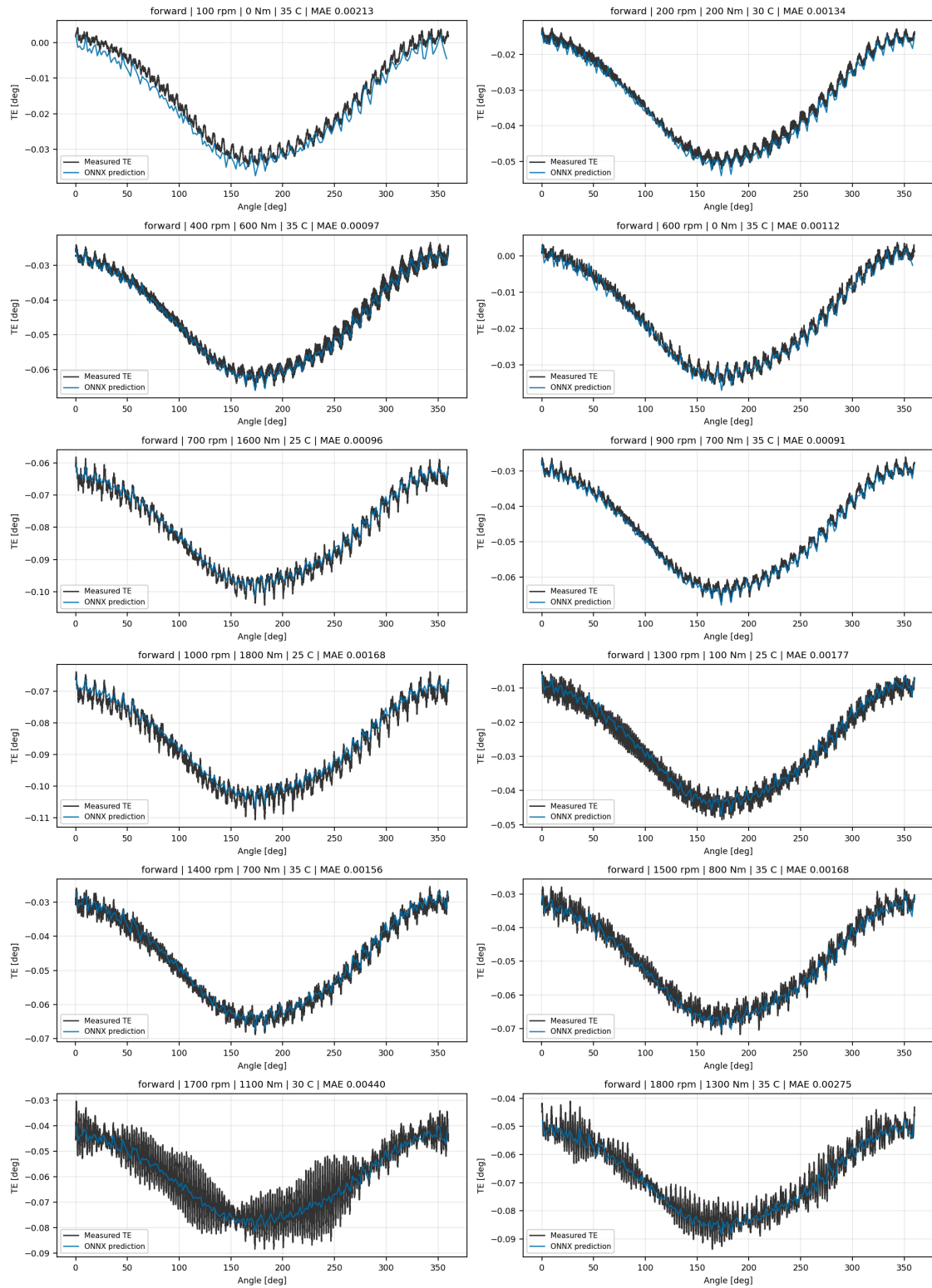
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.001962	0.002342	4.317	9.740
backward	94	0.002561	0.003018	4.717	10.769
global	194	0.002282	0.002718	4.574	10.784

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000112	0.000944	0.001623	0.003539
backward	0.000234	0.000941	0.002181	0.006428
global	0.000058	0.000942	0.001939	0.005049

Forward 12-Curve Page

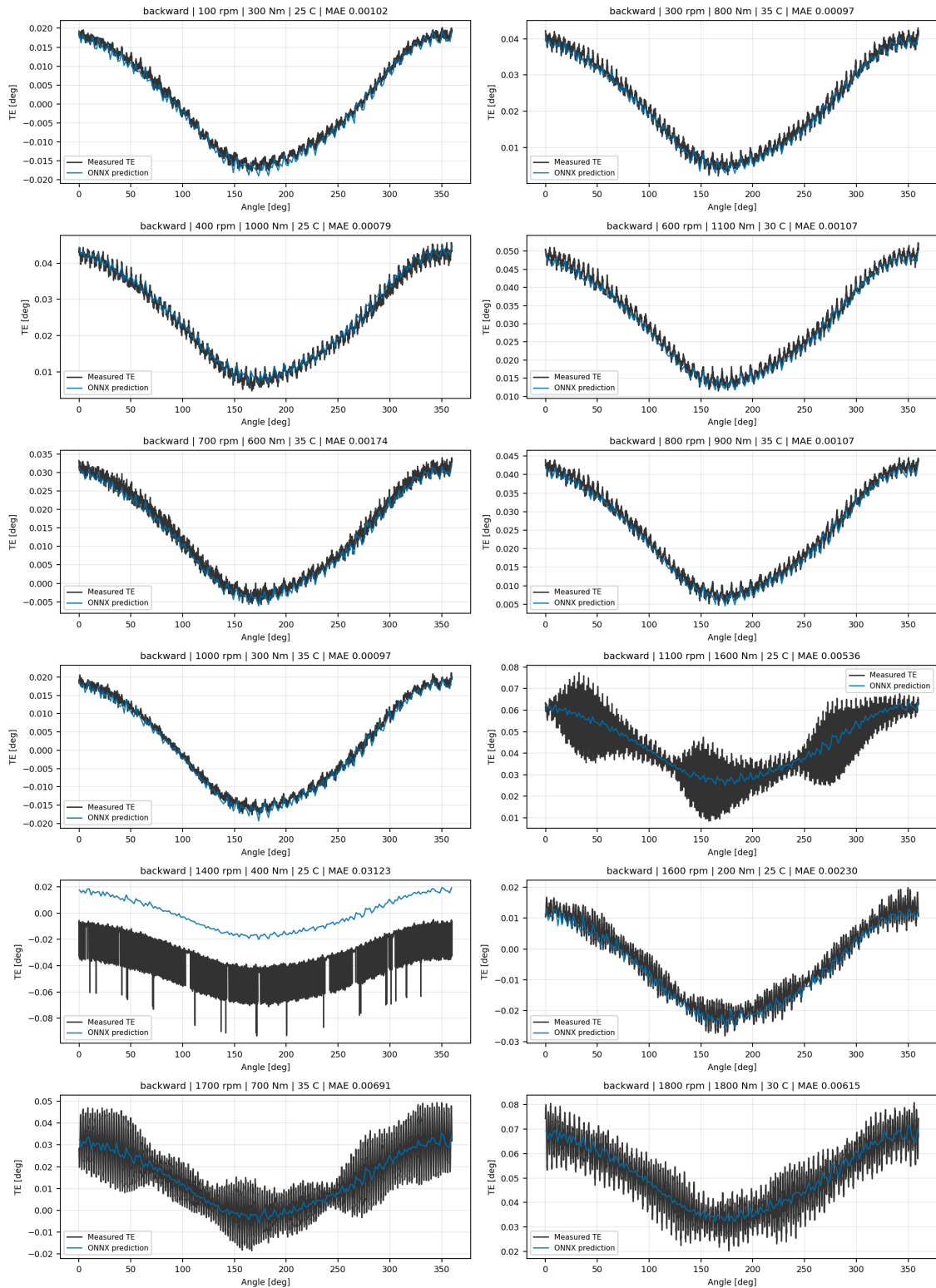
Polished Dataset + Setpoints | residual_harmonic_lstm_sequence_dense360 | forward



polished_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

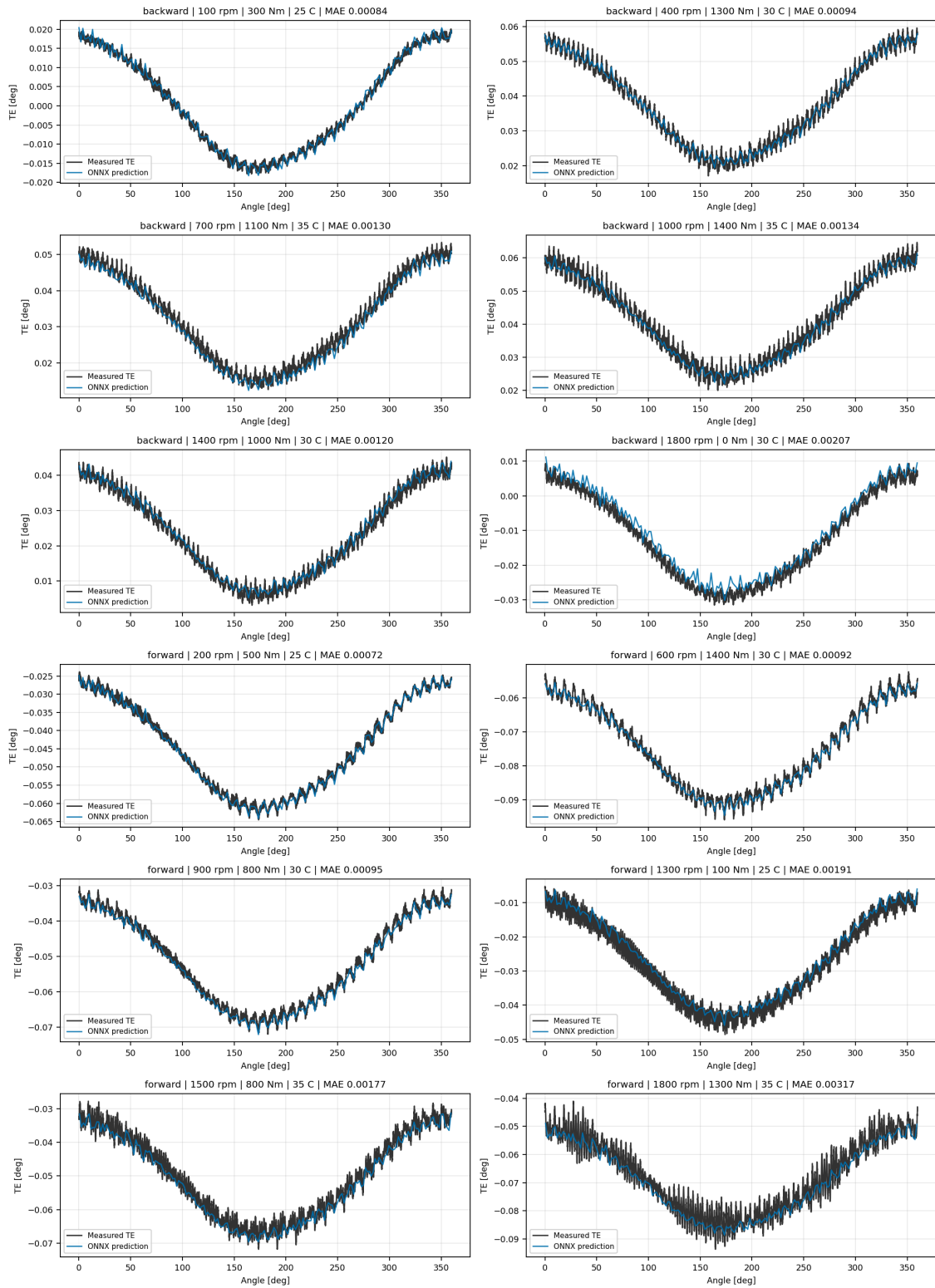
Polished Dataset + Setpoints | residual_harmonic_lstm_sequence_dense360 | backward



polished_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Setpoints | residual_harmonic_lstm_sequence_dense360 | global



polished_dataset_setpoints global 12-curve collage

Polished Dataset + Actual Values

- dataset: polished_dataset ;
- input mode: actual_values ;
- evaluated family: residual_harmonic_lstm_sequence_dense360 ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_residual_harmonic_lstm_sequence_dense360_fw_polished_actual_values	2026-07-10-06-59-39_te_residual_harmonic_lstm_sequence_dense360_fw_polished_actual_values	polished_point_v1
backward	te_residual_harmonic_lstm_sequence_dense360_bw_polished_actual_values	2026-07-10-07-24-50_te_residual_harmonic_lstm_sequence_dense360_bw_polished_actual_values	polished_point_v1
global	te_residual_harmonic_lstm_sequence_dense360_global_polished_actual_values	2026-07-10-06-44-07_te_residual_harmonic_lstm_sequence_dense360_global_polished_actual_values	polished_point_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/actual_values/exported/residual_harmonic_lstm_sequence_dense360/forward/2026-07-10-06-59-39_te_residual_harmonic_lstm_sequence_dense360_fw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/residual_harmonic_lstm_sequence_dense360/forward/2026-07-10-06-59-39_te_residual_harmonic_lstm_sequence_dense360_fw_polished_actual_values/python/residual_harmonic_lstm_sequence-epoch=062-val_mae=0.00203211.ckpt
backward	models/polished_dataset/actual_values/exported/residual_harmonic_lstm_sequence_dense360/backward/2026-07-10-07-24-50_te_residual_harmonic_lstm_sequence_dense360_bw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/residual_harmonic_lstm_sequence_dense360/backward/2026-07-10-07-24-50_te_residual_harmonic_lstm_sequence_dense360_bw_polished_actual_values/python/residual_harmonic_lstm_sequence-epoch=105-val_mae=0.00199856.ckpt
global	models/polished_dataset/actual_values/exported/residual_harmonic_lstm_sequence_dense360/global/2026-07-10-06-44-07_te_residual_harmonic_lstm_sequence_dense360_global_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/residual_harmonic_lstm_sequence_dense360/global/2026-07-10-06-44-07_te_residual_harmonic_lstm_sequence_dense360_global_polished_actual_values/python/residual_harmonic_lstm_sequence-epoch=018-val_mae=0.00209241.ckpt

Aggregate Metrics

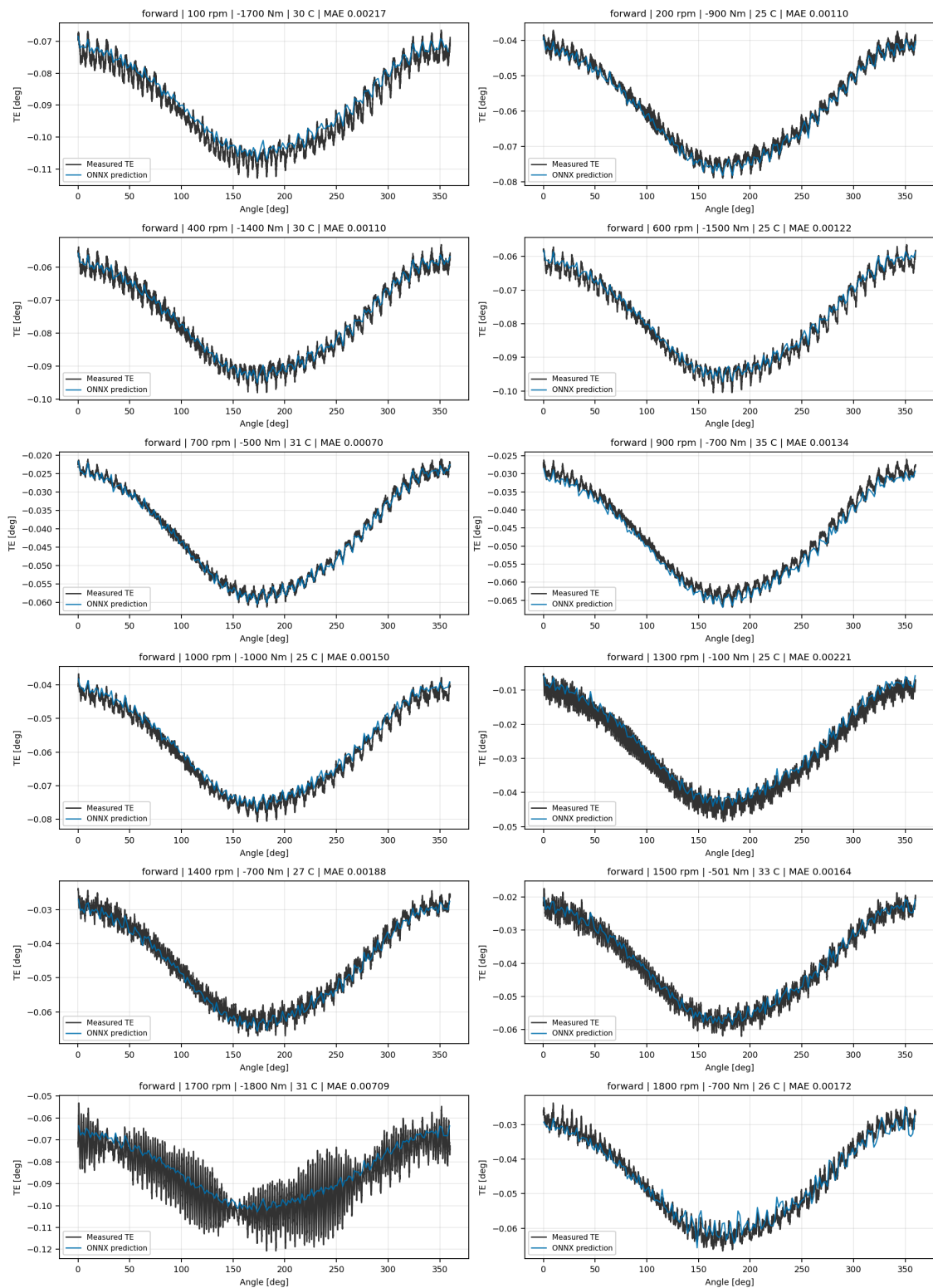
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.001963	0.002351	4.308	9.765
backward	94	0.002407	0.002910	4.655	10.621
global	194	0.002330	0.002768	4.681	10.888

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000074	0.000820	0.001683	0.004364
backward	-0.000045	0.000668	0.002267	0.005942
global	0.000269	0.000925	0.001985	0.005341

Forward 12-Curve Page

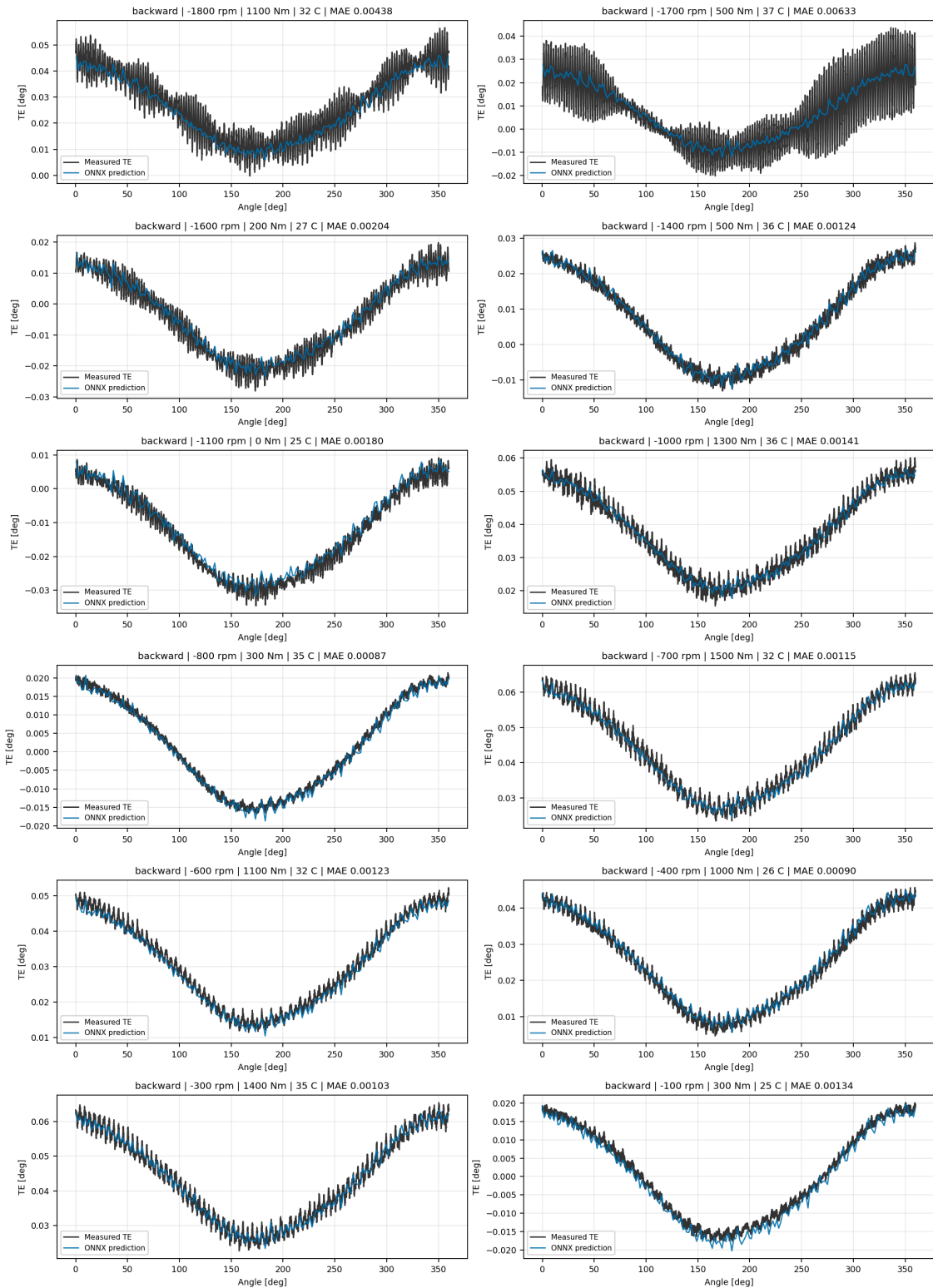
Polished Dataset + Actual Values | residual_harmonic_lstm_sequence_dense360 | forward



polished_dataset_actual_values forward 12-curve collage

Backward 12-Curve Page

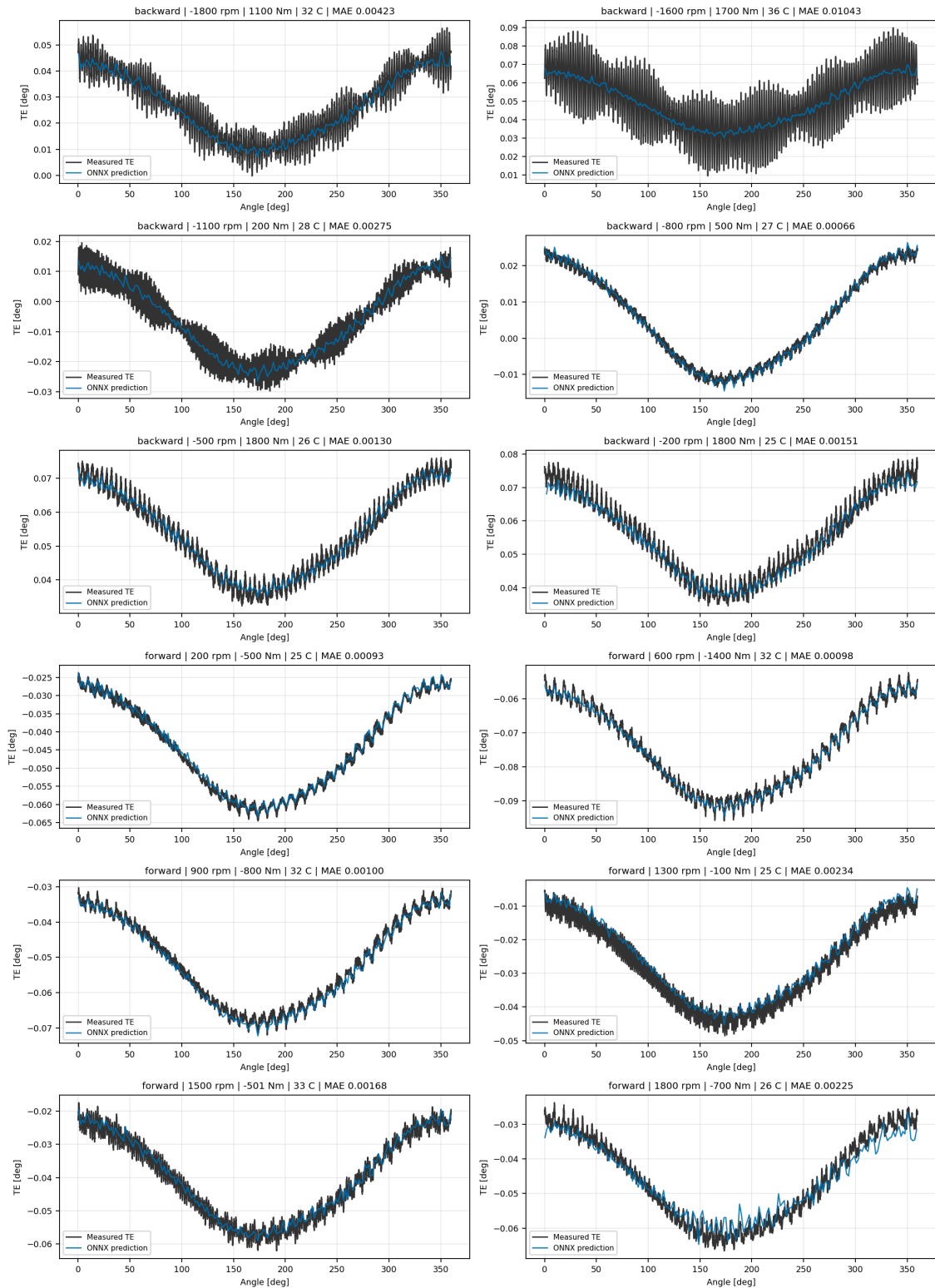
Polished Dataset + Actual Values | residual_harmonic_lstm_sequence_dense360 | backward



polished_dataset_actual_values backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Actual Values | residual_harmonic_lstm_sequence_dense360 | global



polished_dataset_actual_values global 12-curve collage